

AccessMatrix Azure Key Vault HSM Implementation Guide - 6.0.1-GA

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1. Preface - AccessMatrix Azure Key Vault HSM Implementation Guide

[Azure Key Vault](#) is a cloud service that works as a secure secrets store to safeguard encryption keys and secrets such as certificates and passwords. It allows you to host encryption keys and perform cryptographic operations securely. Azure Key Vault can be either:

- Software
- Hardware Security Module (HSM), which allows you to host encryption keys and perform cryptographic operations securely in FIPS 140-2 Level 2 validated HSMs

Intended Reader

The readers must possess some familiarity with Azure concepts like the Azure accounts and others, and operations that include creating common Azure cloud services like Linux VM, virtual networks and service principal.

Document Scope

The scope of this document is to describe how to integrate AccessMatrix with Azure Key Vault. While this document provides necessary steps to create a key vault, it will not provide documentation on operating the Azure that already has its own vast and complete documentation. However, where it is needed, this document will provide:

- Links to Azure documentation to provision/manage services (e.g. VMs, key vaults) that must be prepared for this integration.
- Azure CLI (Command-Line Interface) command-line instruction for resources that are directly used by AccessMatrix (e.g. key vault). While you can manage and configure the key vaults by using various ways (e.g. Azure Portal, Azure PowerShell, Azure CLI, [REST APIs](#)), in this documentation, the instructions are provided by using Azure CLI command.

For any feedback or questions regarding this document, contact doc@i-sprint.com.

Related Documents

For more information, see the following documents in the AccessMatrix Product Suite:

- [AccessMatrix Common Administration Guide](#)

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- [*AccessMatrix Common Deployment Guide*](#)
 - [*UAS Pseudo E2EEA Implementation Guide*](#)

2. Introduction to Azure Key Vault HSM Implementation with AccessMatrix

This document shows you how to implement and integrate AccessMatrix with Azure Key Vault (Cloud HSM).

Terminology

This table lists some common terms that are used in this document. It is not intended as a comprehensive definition for each item.

Term	Description
Azure Command-line Interface (CLI)	Command-line interface installed on the workstation that provides a set of commands used to create and manage Azure resources.
Key Vault	An Azure service to store keys, secrets and certificates securely on Azure cloud, that are cryptographically protected by software or FIPS 140-2 Level 2 validated HSMs.
Key	It is your organization's cryptographic key that will be named and registered in Azure Key Vault.
Service Principal	It is referring to the application Id registration that is used to integrate with Azure Key Vault. In this case, the AccessMatrix application.
Tenant	Tenant is a representation of an organization or Azure customer who would request permissions from Azure Key Vault to carry out the cryptographic operations.
Client Secret	It is the application secret that you generated for your application in Azure App Registration portal.
Key wrapping	It uses a KEK (Key Encryption Key) to encrypt a key for transport or storage.
Key Provider	It is a key manager in AccessMatrix to define communication interface with Azure Key Vault HSM for different implementations of cryptography key provisioning.

AccessMatrix Integration with Azure Key Vault

Azure Key Vault only supports asymmetric cryptographic keys and operations. AccessMatrix uses Azure Key Vault as a key provider, which means it uses Key Vault to store cryptographic keys and perform cryptographic operations (encryption/decryption) inside it. AccessMatrix integrates with Azure Key Vault

as a key provider to host RSA asymmetric keys (for key wrapping) and perform asymmetric cryptographic operations inside Azure Key Vault.

AccessMatrix does not provide key management to Key Vault, e.g. creating the keys, destroying the keys, rotating the keys, etc. All of those key management/operations are to be done directly on Azure Key Vault service (either by using Azure Portal or Azure CLI).

Authentication to Access Azure Key Vault

Any entity or principal that accesses any Azure services must be identified and authenticated.

An Azure subscriber uses credentials comprising an account name, and a key by storing them as Azure Key Vault secrets for authentication purpose. Any entity or service principal (application id) that accesses any Azure Key Vault resources must be identified and authenticated.

- **Identification**

AccessMatrix makes its identity configuration known to Microsoft Entra ID before run-time by registering it in an [Microsoft Entra ID tenant](#).

- **Authentication**

Azure Key Vault supports Microsoft Entra ID access tokens that may be obtained using OAuth2. For more information on registering your application and authenticating to use Azure Key Vault, see [Register your client application with Microsoft Entra ID](#).

Out of Integration Scope

While they might be mentioned in other parts of this document, these items are not part of this integration scope, i.e.

- Key management to Key Vault, e.g. creating the keys, destroying the keys, rotating the keys, etc.
- Symmetric keys and operations. As of the release of this version, Azure Key Vault only supports asymmetric keys.

Implementation Checklist

See below a list of items that must be prepared before integrating AccessMatrix with Azure Key Vault.

Item	Description
Azure Account	You have created an account as an Azure subscriber before you begin to set and retrieve a secret key from Azure Key Vault using Azure CLI.
Azure Key Vault	You have assigned a vault owner who can create an Azure Key Vault and gain full access and control over it.
Key Vault secret	You have created your company secret key. If you have not done so, refer to the Add a secret to Key Vault section on this page .
Application Id (registration)	You have been assigned with an application id during your registration in Azure portal. See Create a Service Principal for more information.
Tenant Id	You have set up a tenant that represents your company to request permissions from Azure Key Vault for cryptographic operations. See Create a Resource Group - Multi-Tenant Supports and Service Principal accesses to Key Vault for more information.
AccessMatrix	You have set up the above items on Azure Key Vault. Next, deploy Azure Key Vault Key Provider Plugin on the AccessMatrix server.



Notes:

- Anyone trying to manage or retrieve content from the Azure Key Vault must be authenticated by Microsoft Entra ID.
 - Authentication establishes the identity caller.
 - Authorization determines which operations the caller can perform. Authorization in Key Vault uses a combination of [Role-based access control](#) (RBAC) and the Azure Key Vault access policies. Define the [custom roles for Azure resources](#) for the specific needs of your organization.
- Control access to Key Vault data by granting permissions for keys, secrets and certificates to the subscribers, resource groups and others. To set access policies for a key vault, use the Azure portal, Azure CLI, Azure PowerShell or the Key Vault Management REST APIs.

Implementation Planning

Security Consideration

Azure Key Vault service only provides asymmetric keys and cryptographic operations (RSA and EC), which are secure but much slower compared with symmetric encryption. AccessMatrix integrates with Azure Key Vault to encrypt/wrap encryption keys in AccessMatrix.

For example, you create PIN Encryption Key (to secure PIN) in AccessMatrix default system key provider and secure it by wrapping it with Azure Key Vault keys.

For additional security consideration, refer to [Security Recommendations for Azure Key Vault](#) for more information.

Performance Consideration

Performance in cryptographic operations is fast because AccessMatrix only calls Azure Key Vault API once to unwrap the encryption keys (for example PIN Encryption Key).

3. Azure Key Vault Service

When you create a key vault in an Azure subscription, it is automatically associated with the Microsoft Entra ID tenant of the subscription. All callers (users or applications) must register properly in this tenant and authenticate to access the key vault. AccessMatrix accesses Key Vault on behalf of a signed-in user.

The following show you how to set and retrieve a key vault and client secret from Azure Key Vault using [Azure Cloud Shell](#) for authentication. You need to create an instance of the KeyClient class that requires a key vault URL and the client's secret credentials like client key name, client Id, and tenant Id to instantiate a client object using the default Azure credential.

This section assumes the prerequisite checklist has been prepared, i.e.

- [Azure Subscription](#).
- [Azure CLI](#) or [Azure PowerShell](#).

Create a Resource Group: Multi-Tenant Supports

An Azure resource group is a logical container into which Azure resources are deployed and managed. Refer to [multi-tenant support](#) to create a resource group. To install the Azure package, for example:

```
{
<dependency>
  <groupId>com.azure</groupId>
  <artifactId>azure-security-keyvault-keys</artifactId>
  <version>4.0.0</version>
</dependency>
<dependency>
  <groupId>com.azure</groupId>
  <artifactId>azure-identity</artifactId>
  <version>1.0.0</version>
</dependency>
}
```

By using Azure CLI, create the resource group, for example:

```
{az group create --name "myResourceGroup" -l "southeastasia"
az keyvault create --name <your-unique-keyvault-name> -g "myResourceGroup"}
```

Create a Service Principal

An Azure service principal is a security identity such as user identity (username and password or certificate) with a specific role, and tightly controlled permissions. You can create a service principal

using the Azure CLI `az ad sp create-for-rbac` command, for example:

```
az ad sp create-for-rbac -n "http://mySP" ~~~-sdk-auth~~~
```

You can also use the Azure Portal, for example:

```
sample@Azure:$ az ad sp create-for-rbac -n "sample-cli" ~~~-sdk-auth
Changing "sample-cli" to a valid URI of "http://sample-cli", which is the required
format used for the service principal names
Creating a role assignment under the scope of "/subscriptions/ff9f4ef7-c33b-454c-9844-
429361615309"
  Retrying role assignment creation: 1/36
  Retrying role assignment creation: 2/36
```

This operation will return a series of key/value pairs.

```
{
  "clientId": "7da18cae-779c-41fc-992e-0527854c6583",
  "clientSecret": "b421b443-1669-4cd7-b5b1-394d5c945002",
  "subscriptionId": "443e30da-feca-47c4-b68f-1636b75e16b3",
  "tenantId": "35ad10f1-7799-4766-9acf-f2d946161b77",
  "activeDirectoryEndpointUrl": "https://login.microsoftonline.com",
  "resourceManagerEndpointUrl": "https://management.azure.com/",
  "activeDirectoryGraphResourceId": "https://graph.windows.net/",
  "sqlManagementEndpointUrl": "https://management.core.windows.net:8443/",
  "galleryEndpointUrl": "https://gallery.azure.com/",
  "managementEndpointUrl": "https://management.core.windows.net/"
}
```



Note: Take note of the client Id, client Secret and tenant Id to be used for access to your key vault.

Create an Azure Key Vault

You can use a key vault to store keys and secrets. This information is stored in a key vault with your key name. AccessMatrix does not need to persist the keys and secrets. The administrator can request them from the key vault as needed. Create an Azure Key Vault by using the Azure CLI, for example:

```
sample@Azure:$ az keyvault create ~~~-name "sample-Vault" ~~~-resource-group
"sampleGroup" ~~~-location southeastasia
{
  "id": "/subscriptions/ff9f4ef7-c33b-454c-9844-
429362615308/resourceGroups/sampleGroup/providers/Microsoft.KeyVault/vaults/sample-
Vault",
  "location": "southeastasia",
  "name": "sample-Vault",
  "properties": {
    "accessPolicies": [
      {
        "applicationId": null,
        "objectId": "ead1397e-4739-4f46-84a7-9d15f0324192",
        "permissions": {
```

```
    "certificates": [
      "get",
      "list",
      "delete",
      "create",
      "import",
      "update",
      "managecontacts",
      "getissuers",
      "listissuers",
      "setissuers",
      "deleteissuers",
      "manageissuers",
      "recover"
    ],
    "keys": [
      "get",
      "create",
      "delete",
      "list",
      "update",
      "import",
      "backup",
      "restore",
      "recover"
    ],
    "secrets": [
      "get",
      "list",
      "set",
      "delete",
      "backup",
      "restore",
      "recover"
    ],
    "storage": [
      "get",
      "list",
      "delete",
      "set",
      "update",
      "regeneratekey",
      "setsas",
      "listsas",
      "getsas",
      "deletesas"
    ]
  },
  "tenantId": "e13fdd6a-4b47-4ce3-9bcd-771d54a3e69d"
}
],
"createMode": null,
"enablePurgeProtection": null,
"enableSoftDelete": null,
"enabledForDeployment": false,
"enabledForDiskEncryption": null,
"enabledForTemplateDeployment": null,
"networkAcls": null,
"provisioningState": "Succeeded",
```

```

    "sku": {
      "name": "standard"
    },
    "tenantId": "e13fdd6a-4b47-4ce3-9bcd-771d54a3e88d",
    "vaultUri": "https://sample-vault.vault.azure.net/"
  },
  "resourceGroup": "sampleGroup",
  "tags": {},
  "type": "Microsoft.KeyVault/vaults"
}

```

Create Service Principal: App Registration in Azure Portal

```

sample@Azure:$ az ad sp create-for-rbac -n "sample-cli" --sdk-auth
Changing "sample-cli" to a valid URI of "http://sample-cli", which is the required
format used for service principal names
Creating a role assignment under the scope of "/subscriptions/ff9f4ef7-c33b-454c-9844-
429361615308"
  Retrying role assignment creation: 1/36
  Retrying role assignment creation: 2/36
{
  "clientId": "852889a4-4c2a-4e0d-9888-39b5ff143893",
  "clientSecret": "e244e6d8-3e51-4a8f-b9ba-f2ae5e288929",
  "subscriptionId": "ff9f4ef7-c33b-454c-9844-429361615308",
  "tenantId": "e13fdd6a-4b47-4ce3-9bcd-771d54a3e68d",
  "activeDirectoryEndpointUrl": "https://login.microsoftonline.com",
  "resourceManagerEndpointUrl": "https://management.azure.com/",
  "activeDirectoryGraphResourceId": "https://graph.windows.net/",
  "sqlManagementEndpointUrl": "https://management.core.windows.net:8443/",
  "galleryEndpointUrl": "https://gallery.azure.com/",
  "managementEndpointUrl": "https://management.core.windows.net/"
}

```

Service Principal Accesses to Key Vault

Next, you allow the service principal accesses to your Key Vault, for example:

```

sample@Azure:$ az keyvault set-policy -n sampleVault --spn 852889a4-4c2a-4e0d-9888-
39b5ff143893 --secret-permissions delete
get list set --key-permissions create decrypt delete encrypt get list unwrapKey
wrapKey
{
  "id": "/subscriptions/ff9f4ef7-c33b-454c-9844-
429361615308/resourceGroups/sampleGroup/providers/Microsoft.KeyVault/vaults/sample-
Vault",
  "location": "southeastasia",
  "name": "sample-Vault",
  "properties": {
    "accessPolicies": [
      {
        "applicationId": null,
        "objectId": "ead1397e-4739-4f46-84a7-9d15f0304192",
        "permissions": {
          "certificates": [
            "get",

```

```
        "list",
        "delete",
        "create",
        "import",
        "update",
        "managecontacts",
        "getissuers",
        "listissuers",
        "setissuers",
        "deleteissuers",
        "manageissuers",
        "recover"
    ],
    "keys": [
        "get",
        "create",
        "delete",
        "list",
        "update",
        "import",
        "backup",
        "restore",
        "recover"
    ],
    "secrets": [
        "get",
        "list",
        "set",
        "delete",
        "backup",
        "restore",
        "recover"
    ],
    "storage": [
        "get",
        "list",
        "delete",
        "set",
        "update",
        "regeneratekey",
        "setsas",
        "listsas",
        "getsas",
        "deletesas"
    ]
},
"tenantId": "e13fdd6a-4b47-4ce3-9bcd-771d54a3e68d"
},
{
    "applicationId": null,
    "objectId": "e5e55d2f-c316-4b0c-a492-23a43ecce55a",
    "permissions": {
        "certificates": null,
        "keys": [
            "create",
            "decrypt",
            "delete",
            "encrypt",
            "get",
```

```

        "list",
        "unwrapKey",
        "wrapKey"
    ],
    "secrets": [
        "delete",
        "get",
        "list",
        "set"
    ],
    "storage": null
},
"tenantId": "e13fdd6a-4b47-4ce3-9bcd-771d54a3e68d"
},
{
    "applicationId": null,
    "objectId": "6d584360-23e5-441a-b435-ee700947cfb0",
    "permissions": {
        "certificates": null,
        "keys": [
            "create",
            "decrypt",
            "delete",
            "encrypt",
            "get",
            "list",
            "unwrapKey",
            "wrapKey"
        ],
        "secrets": [
            "delete",
            "get",
            "list",
            "set"
        ],
        "storage": null
    },
    "tenantId": "e13fdd6a-4b47-4ce3-9bcd-771d54a3e68d"
}
],
"createMode": null,
"enablePurgeProtection": null,
"enableSoftDelete": null,
"enabledForDeployment": false,
"enabledForDiskEncryption": null,
"enabledForTemplateDeployment": null,
"networkAcls": null,
"provisioningState": "Succeeded",
"sku": {
    "name": "standard"
},
"tenantId": "e13fdd6a-4b47-4ce3-9bcd-771d54a3e68d",
"vaultUri": "https://sample-vault.vault.azure.net/"
},
"resourceGroup": "sampleGroup",
"tags": {},
"type": "Microsoft.KeyVault/vaults"
}

```

4. AccessMatrix Azure Key Vault Key Provider Deployment

This part assumes AccessMatrix has been deployed properly on Azure Key Vault. You can confirm it by opening Admin Console, visit *http://<hostname:port>/am5* (or *https://<hostname:port>/am5*).

Deploy Azure Key Vault Key Provider Plugin

Azure Key Vault Key Provider is implemented as a plug-in. Thus, you need to deploy the plug-in into the AccessMatrix folder by following these steps:

1. Copy **am-azurekvkprov-core.jar** file from `<ACMATRIX_ROOT>\sdk\azurekvkprov\` folder to `<ACMATRIX_ROOT>\tomcat\webapps\am5\WEB-INF\lib` folder.
2. Copy `<ACMATRIX_ROOT>\sdk\azurekvkprov\lib\*.jar` files to the `<ACMATRIX_ROOT>\tomcat\webapps\am5\WEB-INF\lib` folder.
3. Copy **am-azurekvkprov-plugin.jar** file from `<ACMATRIX_ROOT>\sdk\azurekvkprov\` folder to `<ACMATRIX_ROOT>\tomcat\plugins` folder.
4. Copy `field-definitions-azurekvkprov.xml` from `<ACMATRIX_ROOT>\sdk\azurekvkprov\res\` folder to `<ACMATRIX_ROOT>\tomcat\webapps\am5\WEB-INF\classes\` folder.
5. Copy **wpecustom-azurekvkprov.properties** from `<ACMATRIX_ROOT>\sdk\azurekvkprov\res\` folder to `<ACMATRIX_ROOT>\tomcat\webapps\am5\WEB-INF\classes\` folder.
6. Rename **wpecustom-azurekvkprov.properties** in the folder `<ACMATRIX_ROOT>\tomcat\webapps\am5\WEB-INF\classes\` to **wpecustom.properties**.
7. Add the following line to the file **amsystem.properties** in the folder `<ACMATRIX_ROOT>\tomcat\webapps\am5\WEB-INF\classes\`:

Properties (.properties files)

```
am.field_definition.file.0=field-definitions-  
azurekvkprov.xml
```

8. Start AccessMatrix server, check **am.log** file (`<hostname>_am.log`) to see if Azure Key Vault Key Provider plugin is loaded. Find this keyword "Plugin loaded: AzureKeyVaultKeyProviderPlugin", for example:

```
2019-12-10 14:10:35.038 [main] INFO - (PluginManager.java:331) Loading plugin instance,  
spId=AzureKeyVaultKeyProviderPlugin, version=0.1,  
instance=com.isprint.am.core.impl.ext.plugin.AzureKeyVaultKeyProviderPlugin@275c5251, loadOrder=10  
2019-12-10 14:10:35.038 [main] INFO - (AzureKeyVaultKeyProviderPlugin.java:28) Plugin loaded:  
AzureKeyVaultKeyProviderPlugin - 0.1
```

5. AccessMatrix Azure Key Vault Key Provider and Encryption Key Configuration

This part assumes the AccessMatrix Azure Key Vault Key Provider plugin has been deployed/installed successfully. You can confirm it by navigating to **Administration Dashboard > Security Policy > Key Management > Key Manager > Key Provider**, click the **Create** button, and see if "Azure Key Vault Key Provider" is listed in the Implementation drop-down list.

Besides, you should have completed the steps from the [Azure Key Vault Service](#) section and generated a key name, key vault URL, tenant Id, client Id and client secret to configure the key provider and encryption keys in AccessMatrix to use Azure Key Vault.

Setup Steps

This section shows you how to configure the AccessMatrix server to integrate with Azure Key Vault on the Admin Console. Before you start, it is assumed that you have completed the prerequisites.

- **Configure Azure Key Vault Key Provider**

This key provider uses the client libraries as the crypto library in Azure Key Vault. Use the "Azure Key Vault Key Provider" implementation in AccessMatrix server to integrate with Azure Key Vault. Refer to [Configure Azure Key Vault Key Provider](#) for more details.

- **Configure Wrapper Key**

This asymmetric wrapping key supports the encryption operations by using the public portion of the key to protect and wrap the symmetric keys in the Azure Key Vault. This operation requires the wrap key permission. Refer to [Configure Wrapper Key](#) to learn how to create the wrapper key in the AccessMatrix Admin Console.

- **Configure PIN Encryption Key (PEK)**

PEK (PIN Encryption Key) is the key that uses AES (Advanced Encryption Standard) to encrypt the PIN in the AccessMatrix security registry. PEK is wrapped by Azure wrapping key. PEK is randomly generated based on the algorithm provided by the JCE (Java Cryptographer Extension). Refer to [Configure PIN Encryption Key \(PEK\)](#) to learn how to use Azure wrapper key to protect this Pin Encryption Key.



Notes:

- Refer to [AccessMatrix Common Administration Guide - Configure Encryption Key](#), for the explanation on each key's purpose.

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- If the use case is for PseudoE2EEA, refer to [UAS Pseudo E2EEA Implementation Guide](#).

Configure Azure Key Vault Key Provider

Upon successful deploying Azure Key Vault Key Provider plugin on the AccessMatrix server, the Azure Key Vault Key Provider will be created by default in the Admin Console.

1. Navigate to **Administration Dashboard > Security Policy > Key Management > Key Manager > Key Provider** on the left pane.
2. Click **Create** and select "Azure Key Vault Key Provider" from the **Implementation** drop-down list.
3. Click **Next**.
4. Enter a **Key Provider Id** (e.g. "AZKeyProv") and a **Key Provider Name** (e.g. "AZKeyProv") and enter its description.
5. Specify the **Directory (tenant) ID** (e.g. "e13fdd6a-4b47-4ce3-9bcd-771d54a3e78d").
6. Specify the **Application (client) ID** (e.g. "ec359610-7662-4d37-a708-abe69a678ffb").
7. Specify the **Azure Key Vault URL** (e.g. "https://sample-vault.vault.azure.net/").
8. Specify the **Client Secret** (e.g. "t/Ri3-SzHGlV6TYr=edO/tcTj4S77uU7").
9. Optionally, if a proxy server is required, specify the following settings under the HTTP Proxy header:
 - a. **Enabled** - Setting this to "true" will enable the HTTP Proxy.
 - b. **IP address** - Refers to the IP address of the proxy server.
 - c. **Port** - Refers to the port of the proxy server.
 - d. **User Name** - Refers to the username used to authenticate the proxy server.
 - e. **User Password** - Refers to the user's password used to authenticate the proxy server.
10. Click **Save**. A new Key Provider is created.

Configure Wrapper Key

i Note: This is a wrapping key stored in Azure Key Vault that is used to protect/wrap an encryption key (i.e. PEK - Pin Encryption Key). It is assumed that you have already created the key in Azure Key Vault before proceeding with the following steps.

-
1. Navigate to **Administration Dashboard > Security Policy > Key Management > Encryption Key** on the left pane.
 2. Click **Create** and select the newly created key provider (e.g. "AZKeyProv") from the **Key Provider** drop-down list. Notice there is mainly one available **Encryption Key Template**, "Asymmetric key (wrapping key)", as AccessMatrix server integrates with Azure Key Vault to perform asymmetric operations for key encryption.
 3. Click **Next**.
 4. Enter an **Encryption Key Id** (e.g. "asyAzure").
 5. Enter an **Encryption Key Name** (e.g. "asyAzure"), and then enter its description.
 6. In the **Key Name** drop-down list, select the encryption key within Azure Key Vault that will be used as the wrapper key.



Note: You must create the encryption key in Azure Key Vault for it to show up in the drop-down list.

7. Select "Key Wrapping" from the **Purpose** drop-down list under the **Attributes** tab.
8. Click **Save**. A Wrapping key is created.



Note: You can click the **Test KeyInfo** and **View Public Key** buttons to check if this key is created successfully.

Using API

Azure Key Vault uses Azure SDK for Java API to communicate with the AccessMatrix server for authentication. Refer to [Java API Browser](#) for more information. For general information about using Azure REST interfaces, including authentication/authorization and how to acquire an access token, see [Azure REST API Reference](#).
